

# Operating a Volunteer OpenSky Node in Brazil:

## Field Notes on Coverage, Uptime, and Local Regulations

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### KEY RESULTS

189 →  
222 km

#### MEDIAN RANGE

Era A → Era B

+15.6%

#### MEAN RANGE

same 2,234 aircraft

188.5 → 217.9 km; pooled position ranges, paired windows.

99.54%

#### RECEIVER UPTIME

2 confirmed receiver outages  
5 h 12 min, 6 July  
9 h 47 min, 2 August

99.33% reception continuity; 40 gaps > 120 s.

19 start at 05:00 to 07:00 UTC. Receiver statistics cannot distinguish these night gaps from no observable traffic.

786 km

#### MAXIMUM OBSERVED RANGE

3 EVENTS > 600 km

16 to 18 June  
15 August  
25 August

Exact maximum: 786.3 km.  
Outside event windows: 599.9 km.

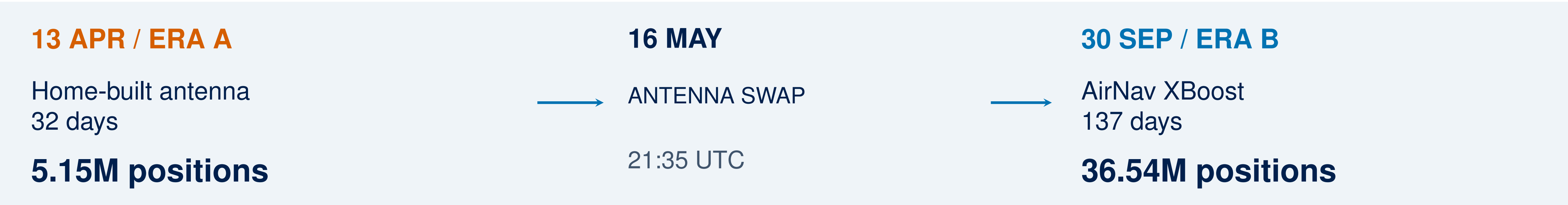
#### COST

US\$150 →  
US\$253  
+69%

Illustrative Remessa Conforme kit:  
US\$150 sticker + US\$60 duty + US\$43 ICMS.  
ICMS 17% gross-up. Total approximate.

### One Brazilian feeder • 169 days of ADS-B observations

Coverage improved after an antenna change; power was not the bottleneck at this Southeast Brazil site.

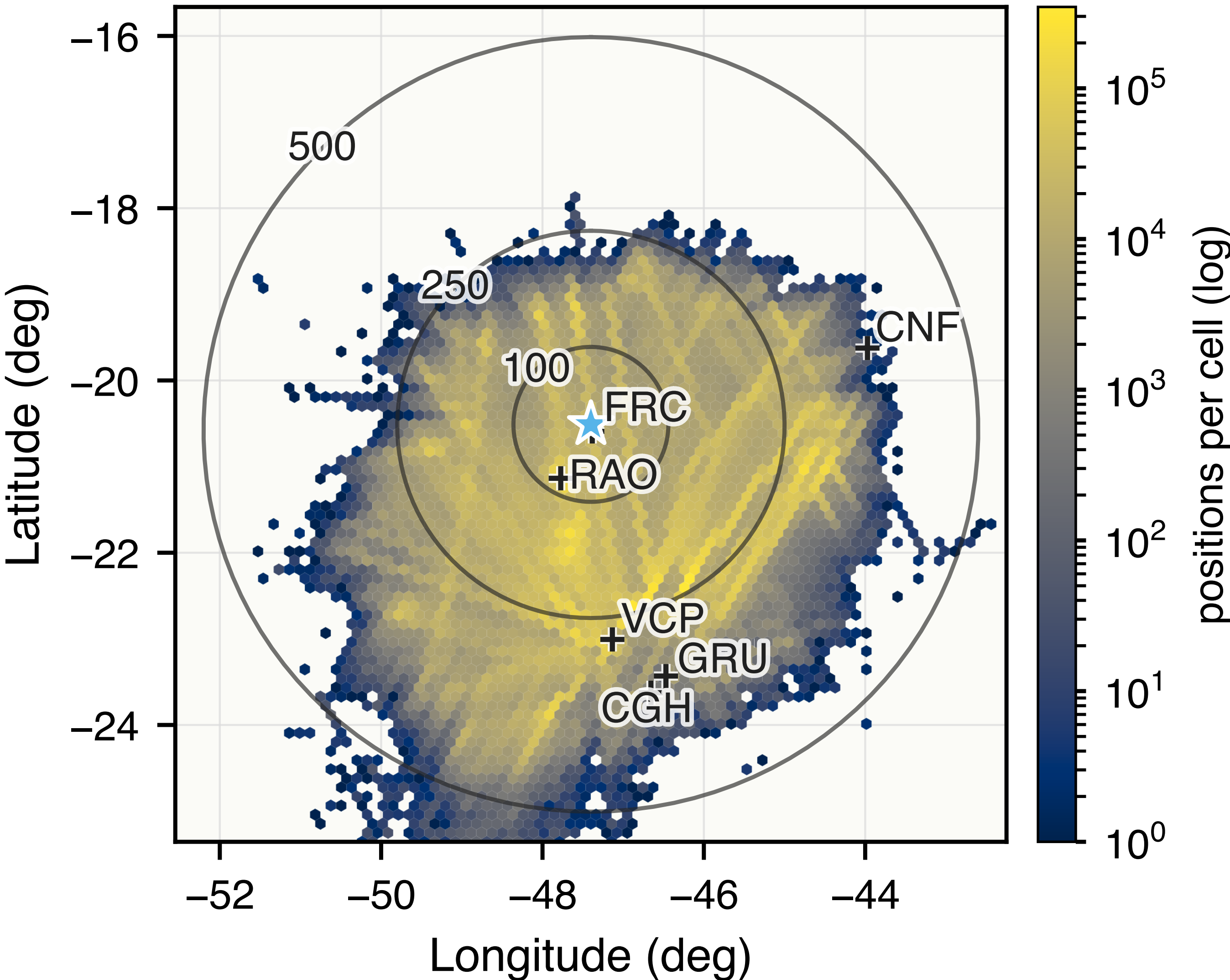


Local readsb traces • direct ADS-B only • MLAT excluded

adsb\_icao / adsb\_icao\_nt Era A: 13 Apr to 14 May; Era B: 16 May to 30 Sep 2026.

One node: local evidence, not a network-wide OpenSky coverage claim.

### COVERAGE What changed after the antenna swap?

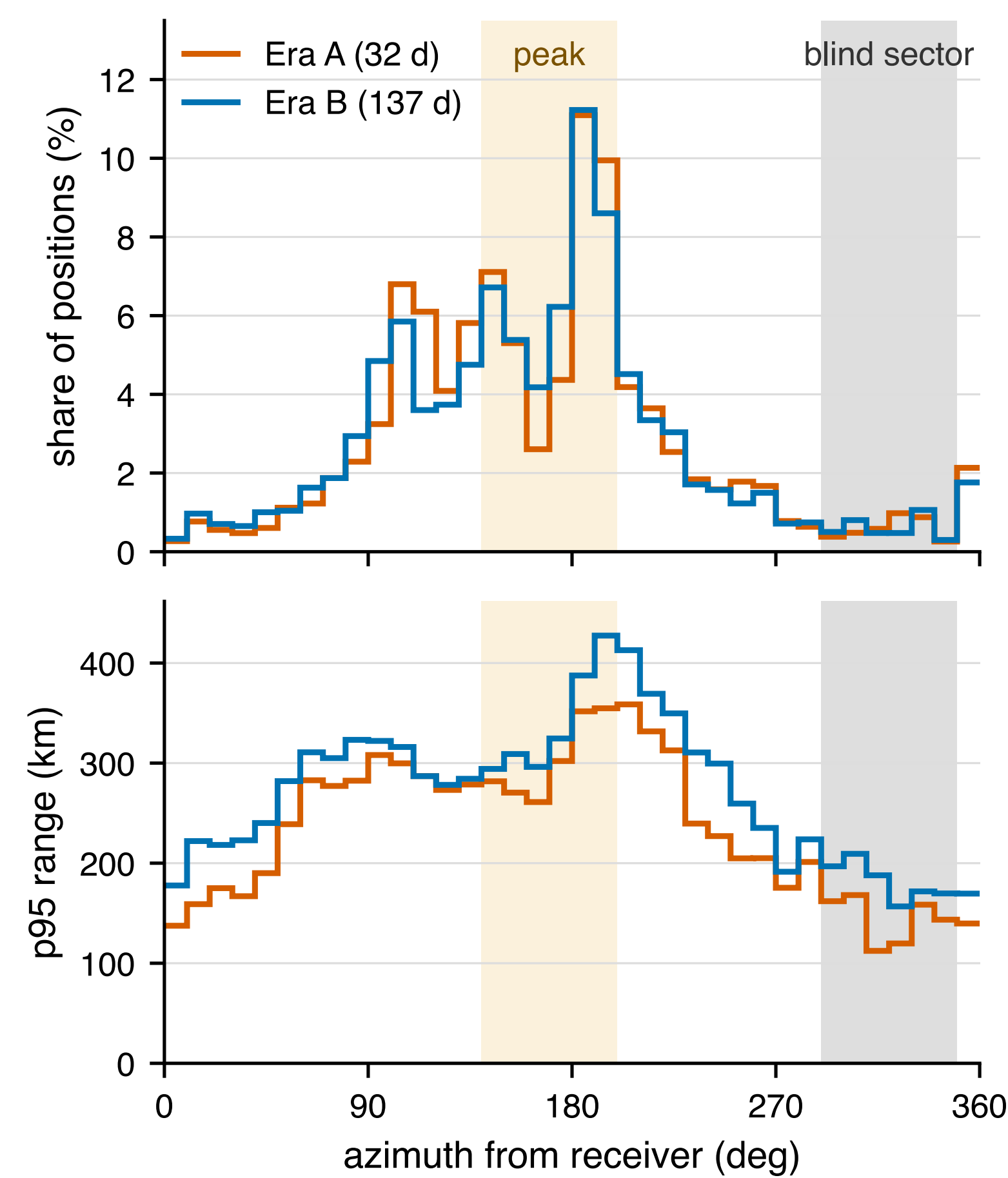


Era B density; rounded (-20.51, -47.40). Rings in km.

| Era      | Positions  | Aircraft | Median | P95   |
|----------|------------|----------|--------|-------|
| A        | 5,153,167  | 2,890    | 189.3  | 299.7 |
| B full   | 36,535,422 | 4,925    | 222.0  | 350.1 |
| B paired | 8,123,970  | 3,159    | 224.0  | 350.6 |

Ranges in km. Paired B: 32 days. Era A max: 510.3 km.

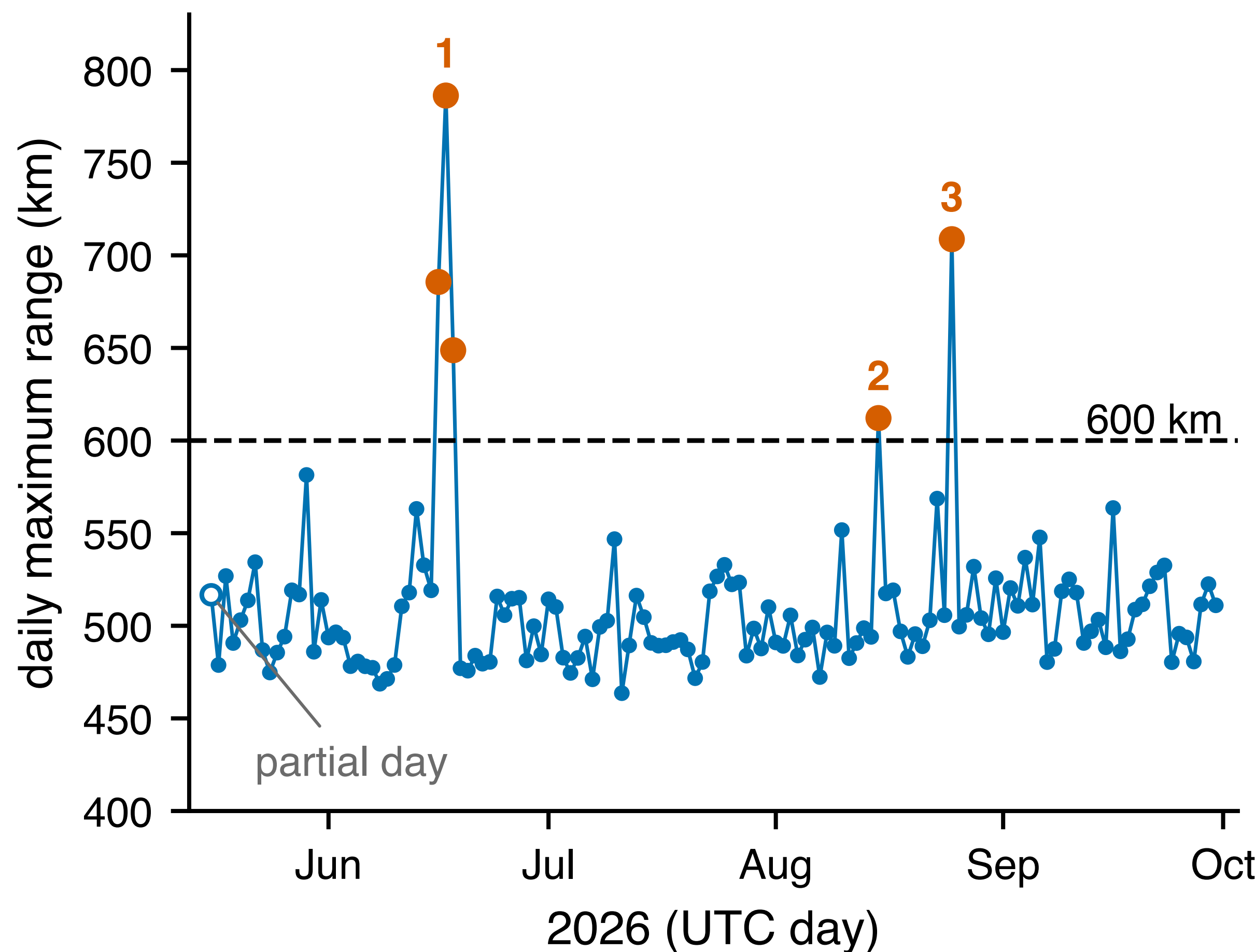
### RANGE BY AZIMUTH



Blind sector 290 to 350° persisted with both antennas, pointing to local terrain/obstruction.

Peak: 140 to 200°, towards the São Paulo terminal area.

### DAILY MAXIMUM RANGE



Context: Idec (2023), ANEEL (2026), ANATEL 715/2019, LGPD, DECEA, MF 1.342/2026. References via QR.

### KEY TAKEAWAYS

| COST                                            | INFORMATION                                              | POWER                                                  |
|-------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------|
| Import taxation amplifies the hardware barrier. | Rules are fragmented rather than explicitly prohibitive. | The constraint is regional, not uniform across Brazil. |

### BRAZILIAN CONTEXT

#### INFORMATION

- ANATEL equipment homologation
- LGPD home-location privacy; international data transfer
- DECEA no continental ADS-B Out mandate

#### POWER Regional context

DEC: interruption hours/consumer, 2022.

Southeast ~7 h / 4 FEC

North ~24 h / 12 FEC

Brazil 2025 DEC: 9.30 h.  
FEC: interruptions/consumer.

Power was not binding at this Southeast node; no national generalization.

FULL PAPER  
DATA • CODE  
REPRODUCIBILITY

Zenodo  
All versions

